Contrastive CUR: Interpretable Joint Feature and Sample Selection for Case-Control Studies

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1 T-TEST AND WILCOXON

1.1 Mouse Protein

Method	% Overlap with CCUR	CCUR-unique Genes
Welch t-test	27%	H3MeK4, pCFOS, SHH, pAKT, Tau
Wilcoxon	27%	H3MeK4, pCFOS, SHH, pAKT, Tau

Table 1. Mouse Protein. Overlap between CCUR and differential expression baselines. We report the percentage of overlapping genes with CCUR along with uniquely identified by CCUR and not selected by each baseline method.

1.2 Small Molecule

Method	% Overlap with CCUR	CCUR-unique Genes
Welch t-test	30%	SAA1, FDXR, LCE1C, KANK3, LCE1F
Wilcoxon	10%	SAA1, FDXR, LCE1C, KANK3, LCE1F

Table 2. Small Molecule. Overlap between CCUR and differential expression baselines. We report the percentage of overlapping genes with CCUR along with uniquely identified by CCUR and not selected by each baseline method.

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1.3 Pathogen

Method	% Overlap with CCUR	CCUR-unique Genes
Welch t-test	20%	Mptx2, Wfdc18, Pla2g4c, Saa1, Retnlb
Wilcoxon	0%	Mptx2, Wfdc18, Pla2g4c, Hspa1a, Retnlb

Table 3. Small Molecule. Overlap between CCUR and differential expression baselines. We report the percentage of overlapping genes with CCUR along with uniquely identified by CCUR and not selected by each baseline method.